

Exp 6-Finite Word length Effects

6.1 Uniform Quantization and Quantization Error Analysis of a Discrete-Time Sinusoidal Signal Using MATLAB

PROGRAM

```
clc;
clear;
close all;
%% Signal
fs = 1000; f = 50;
t = 0:1/fs:0.1;
x = sin(2*pi*f*t);
%% Quantization Levels
L = [8 16 32];
colors = {'r','g','m'};
figure;
for i = 1:length(L)
    % Quantization
    xq{i} = round((x+1)*(L(i)-1)/2)*(2/(L(i)-1)) - 1;
    % Error & MSE
    e{i} = x - xq{i};
    mse(i) = mean(e{i}.^2);
    % Plot Quantized Signals
    subplot(4,1,i+1);
```

```

stairs(t,xq{i},colors{i},'LineWidth',1.2);
title(['Quantized Signal ( ' num2str(L(i)) ' Levels)']);
grid on;
end
% Original Signal
subplot(4,1,1);
plot(t,x,'LineWidth',1.5);
title('Original Signal'); grid on;
% Display MSE
fprintf('MSE for 8 Levels = %f\n', mse(1));
fprintf('MSE for 16 Levels = %f\n', mse(2));
fprintf('MSE for 32 Levels = %f\n', mse(3));
%% Error Plots
figure;
for i = 1:3
    subplot(3,1,i);
    plot(t,e{i});
    title(['Error ( ' num2str(L(i)) ' Levels)']);
    grid on;
end

```

OUTPUT

```

MSE for 8 Levels = 0.010473
MSE for 16 Levels = 0.001763
MSE for 32 Levels = 0.000357

```

